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RESEARCH ARTICLE

Investigation of native isolates of entomopathogenic fungi for the biological control of three citrus pests

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The pathogenicity of 15 isolates of *Beauveria bassiana* (Balsamo) Vuillemin, five isolates of *Metarhizium anisopliae* (Metschnikoff) Sorokin and one isolate of *M. flavoviride* (Gams and Rozsypal) were tested under laboratory conditions against the subterranean life stages of the citrus pests, *Ceratitis rosa* Karsch, *C. capitata* Wiedemann (Diptera: Tephritidae) and *Thaumatotibia leucotreta* Meyrick. (Lepidoptera: Tortricidae). When these citrus pests were treated with a concentration of 1×10^7 conidia mL⁻¹, fungal isolates had a significantly greater effect on the adults of *C. rosa* and *C. capitata* than they did on the puparia of these two fruit fly species. Further, *C. rosa* and *C. capitata* did not differ significantly in their response to entomopathogenic fungi when adult and pupal mycosis were considered. Depending on fungal isolate, the percentage of *T. leucotreta* adults which emerged from fungal treated sand ranged from 5 to 60% and the percentage of pupae with visible signs of mycosis ranged from 21 to 93%. The relative virulence of the four most promising fungal isolates, as well as the commercially available *B. bassiana* product, BroadBand[®] (Biological Control Products, South Africa), were compared against one another as log-probit regressions of mortality against *T. leucotreta* which exhibited a dose-dependent response. The estimated LC₅₀ values of the three most virulent *B. bassiana* isolates ranged from 6.8×10^5 to 2.1×10^6 conidia mL⁻¹, while those of the least pathogenic ranged from 1.6×10^7 to 3.7×10^7 conidia mL⁻¹.

Keywords: *Ceratitis capitata*; *Ceratitis rosa*; *Thaumatotibia leucotreta*; *Beauveria bassiana*; *Metarhizium anisopliae*

1. Introduction

In 2008, the South African citrus industry exported up to 89 million cartons (68% of total citrus production in South Africa) or approximately 1335,000 tons (~618 million US dollars) of fresh fruit (J. Edmonds, personal communication, September 2008). Valencia and Navel oranges are the most popular cultivars on overseas markets with approximately 615,000 tons and 300,000 tons of each exported annually. This is followed by grapefruit (~210,000 tons), lemons (~127,500 tons) and soft citrus (~105,000 tons) (CGA Annual Report 2010). Overseas markets are stringent about the importation of fruit which have been exposed to some pesticides

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and the recent lowering of maximum residue limits for several organophosphates and other chemicals necessitates the investigation of alternative control methods for citrus pests (Grout and Stephen 2005).

Three of the most important citrus pests in South Africa are the false codling moth (FCM), *Thaumatotibia leucotreta* (Meyrick) (Lepidoptera: Tortricidae), Mediterranean fruit fly, *Ceratitis capitata* (Wiedemann) and Natal fruit fly, *Ceratitis rosa* (Karsch) (Diptera: Tephritidae) (Van den Berg 2001; Moore, Grout, Hatting, and Hofmeyr 2008). Fruit flies are regarded as some of the world's most damaging fruit pests due to their highly polyphagous diets and wide habitat tolerances (De la Rosa, Lopez, and Liedo 2002). Fruit flies affect the production of quality fruits by infestation that causes both direct and indirect loss of product. Direct loss occurs through fruit flies feeding on citrus which causes fruit drop and renders the fruit inedible because of fungal infections which invade the fruit after puncture marks are created by ovipositing females (Du Toit 1998). Indirect loss arises from phytosanitary restrictions imposed by importing countries to prevent entry and establishment of fruit flies (Ekesi, Dimbi, and Maniania 2007).

Similarly, false codling moth can cause major economic losses, mainly due to pre-harvest fruit drop and post-harvest decay of fruit caused by the cryptic, internal feeding of the larvae (Kirkman and Moore 2007). Larval feeding and development can affect fruit development at any stage, causing premature ripening and infested fruit may drop as early as November when they are no more than 15–20 cm in diameter (Newton 1998). False codling moth is also considered to be a quarantine pest in the export of citrus fruit to some markets (Stibick 2008). Rejection of fruit during inspection for export to such markets contributes to industry revenue losses associated with this pest (Moore 2002).

To maintain a stronghold in overseas export markets, it has become important for the South African citrus industry to investigate other facets of integrated pest management (IPM), which includes the study and use of microbial control agents, such as entomopathogenic fungi. Well-described entomopathogenic fungal species such as *B. bassiana* and *M. anisopliae* are soil borne and the biology of these three citrus pests includes them dropping to the ground below trees to pupate in the soil (Thomas et al. 2001; Stibick 2008). The development and formulation of control strategies using these fungi as subterranean control agents, targeted at larvae and pupae in the soil, can potentially benefit existing IPM programmes in citrus in South Africa.

The use of entomopathogenic fungi in the form of mycoinsecticides has many desirable traits: fungi leave no toxic residues on crops and pose low or minimal risk to beneficial and other non-target insects (Goettel 1995; Zimmermann 2007a, b). The main impact of entomopathogenic fungi on pest populations however may only occur once economic thresholds have been surpassed. Because chemical control can be really effective at immediately suppressing a growing pest population it is unlikely that entomopathogenic fungi will ever replace chemical control and it is highly unlikely that the disuse of chemicals will occur (Inglis, Goettel, Butt, and Strasser 2001). Thus, it becomes more important to select virulent fungal strains which are pesticide tolerant and environmentally well-adapted for the cropping systems for which they are required. In a previous study various isolates of entomopathogenic fungi were obtained using the *Galleria* bait method from several South African citrus orchards and the natural areas surrounding the citrus orchards (Goble, Dames, Hill,

and Moore 2010). This was undertaken to source potentially virulent and host-specific fungal isolates which may be capable of suppressing key citrus pests. Here we report on the laboratory investigation of pathogenicity of these isolates and their potential as biological control agents.

The aims of the present study were therefore (1) to compare the pathogenicity of 21 native isolates of entomopathogenic fungi against *T. leucotreta*, *C. capitata*, and *C. rosa* final instar larvae using sand-conidial suspension incorporation bioassays, and (2) to test the four most virulent isolates, and a commercially available product BroadBand® (Biological Control Products, South Africa) (which has *B. bassiana* as the active ingredient) against *T. leucotreta* using various concentrations of *B. bassiana* conidia incorporated into sand to ascertain the LC₅₀ and LC₉₀ values of each isolate.

2. Materials and methods

2.1 Insect cultures

Ceratitis capitata and *C. rosa* larvae were obtained from a laboratory culture held at Citrus Research International (CRI), Nelspruit in South Africa, and maintained on an artificial carrot-based diet until late instar larvae were needed. *Thaumatotibia leucotreta* larvae were obtained as late instar larvae from a laboratory culture held at River Bioscience, Addo, South Africa.

2.2 Fungal cultures

The pathogenicity of 15 *B. bassiana*, five *M. anisopliae*, and one *M. flavoviride* isolates (Table 1), obtained using the insect-bait method, were tested in the laboratory against the final instar larvae of *T. leucotreta*, *C. capitata*, and *C. rosa*. All isolates are currently housed in the South African National Collection of Fungi at the Plant Protection Research Institute (PPRI-Agricultural Research Council) in Pretoria, South Africa. Fungal isolates used in this study were grown on Sabouraud Dextrose Agar (SDA) supplemented with 50 mg L⁻¹ Dodine, 50 mg L⁻¹ Chloramphenicol, 50 mg L⁻¹ Ampicillin, or 50 mg L⁻¹ Rifampicin and maintained at 25°C in the dark. Fungal isolates were labelled according to the bait insect species they were originally isolated from, followed by the name of the farm and then the orchard or refugia soil sample number (Goble et al. 2010).

2.3 Preparation of conidial suspensions

Fungal conidia were harvested from 2- to 3-week-old cultures by scraping with a glass rod. Conidia were suspended in 20 mL sterile distilled water supplemented with 0.05% Triton X-100 in sterile glass bottles containing 3-mm glass beads. Bottles were sealed and vortex mixed for 2 min to produce a homogenous conidial suspension. Conidial concentrations were determined using a Neubauer haemocytometer following serial dilution in sterile distilled water and conidial suspensions were used within 3 h of enumeration. Viability of conidia was determined by spread plating 0.1 mL of conidial suspension (titrated to 1 × 10⁷ conidia per mL⁻¹) on three SDA plates. A sterile cover slip was placed on each plate and incubated in complete

Table 1. Twenty-one native entomopathogenic fungal isolates obtained using various bait insects and tested against *T. leucotreta*, *C. capitata*, and *C. rosa* final instar larvae in this study.

Fungal species	Rhodes code	Isolate No.	Host species	Locality	Collection habitat	Soil type	Co-ordinates	Date of isolation
<i>B. bassiana</i>	FCM 11 3 B2	PPRI 9592	<i>T. leucotreta</i>	Mosslands	cultivated orchard soil	Oakleaf Caledon	33°23'54"S; 26°25'41"E	April 2008
	G 14 2 B5	PPRI 9555	<i>G. mellonella</i>	Mosslands	cultivated orchard soil	Oakleaf Richie	33°23'54"S; 26°25'41"E	May 2008
	G Moss R12	PPRI 9593	<i>G. mellonella</i>	Mosslands	virgin soil	Unknown	33°23'54"S; 26°25'41"E	May 2008
	FCM 10 13 L1	PPRI 9680	<i>T. leucotreta</i>	Mosslands	cultivated orchard soil	Oakleaf Caledon	33°23'54"S; 26°25'41"E	April 2008
	G Moss R1	PPRI 9774	<i>G. mellonella</i>	Mosslands	virgin soil	Unknown	33°23'54"S; 26°25'41"E	June 2008
	G Moss R10	PPRI 9557	<i>G. mellonella</i>	Mosslands	virgin soil	Unknown	33°23'54"S; 26°25'41"E	June 2008
	G Moss R7	PPRI 9554	<i>G. mellonella</i>	Mosslands	virgin soil	Unknown	33°23'54"S; 26°25'41"E	May 2008
	G Ar 17 B3	PPRI 9679	<i>G. mellonella</i>	Arundel	cultivated orchard soil	Loam	33°30'57"S; 25°39'11"E	May 2008
	G B Ar 23 B3	RHO Ar 23 B3	<i>G. mellonella</i>	Arundel	cultivated orchard soil	Loam	33°30'57"S; 25°39'11"E	May 2008
	FF J&B R5	PPRI 9556	<i>C. capitata</i>	J&B Citrus	virgin soil	Unknown	32°45'56"S; 25°45'46"E	April 2008
	G J&B 14 L2	PPRI 9678	<i>G. mellonella</i>	J&B Citrus	cultivated orchard soil	Loam	32°45'56"S; 25°45'46"E	June 2008
	G OL R10	PPRI 9691	<i>G. mellonella</i>	Olifantskop	virgin soil	Unknown	33°37'14"S; 25°40'49"E	July 2008
	G OL R11	PPRI 9690	<i>G. mellonella</i>	Olifantskop	virgin soil	Unknown	33°37'14"S; 25°40'49"E	July 2008

Table 1 (*Continued*)

Fungal species	Rhodes code	Isolate No.	Host species	Locality	Collection habitat	Soil type	Co-ordinates	Date of isolation
	G Rose 23 L2	PPRI 9687	<i>G. mellonella</i>	Rosedale	cultivated orchard soil	Oakleaf Limpopo	33°32'21"S; 25°41'39"E	August 2008
	FCM Rose R9	PPRI 9685	<i>T. leucotreta</i>	Rosedale	virgin soil	Unknown	33°32'21"S; 25°41'39"E	August 2008
<i>M. anisopliae</i>	G 14 2 B3	PPRI 9562	<i>G. mellonella</i>	Mosslands	cultivated orchard soil	Oakleaf Richie	33°23'54"S; 26°25'41"E	March 2008
	G 11 3 L6	PPRI 9803	<i>G. mellonella</i>	Mosslands	cultivated orchard soil	Oakleaf Caledon	33°23'54"S; 26°25'41"E	April 2008
	FCM AR 23 B3	PPRI 9561	<i>T. leucotreta</i>	Arundel	cultivated orchard soil	Loam	33°30'57"S; 25°39'11"E	April 2008
	G OL R3	PPRI 9800	<i>G. mellonella</i>	Olifantskop	virgin soil	Unknown	33°37'14"S; 25°40'49"E	July 2008
	G OL R8	PPRI 9801	<i>G. mellonella</i>	Olifantskop	virgin soil	Unknown	33°37'14"S; 25°40'49"E	July 2008
<i>M. flavoviride</i>	G Moss R2	PPRI 9560	<i>G. mellonella</i>	Mosslands	virgin soil	Unknown	33°23'54"S; 26°25'41"E	June 2008

Note: PPRI, Plant Protection Research Institute Fungal Collection accession no.; RHO, Rhodes University accession no.; FCM, False codling moth, *Thaumatotibia leucotreta*; FF, Fruit fly, *Ceratitis capitata*; *Galleria*, *G. mellonella*.

darkness at 25°C. Percentage germination was examined after 24 h from 100-spore counts on each plate. As the pathogenicity of the native fungal isolates was unknown, it was decided that a standard test concentration of 1×10^7 conidia per mL⁻¹ would initially be used against all test insect species. However, a decision was taken to include 1×10^8 conidia per mL⁻¹ concentration against fruit flies. Ekesi, Maniania, and Lux (2002) used a standard test concentration of 1×10^8 conidia per mL⁻¹ against fruit flies in their investigation of Kenyan entomopathogenic fungal isolates.

2.4 Inoculation of insects

Autoclaved, dried river sand (sand particle size: 500 micron) was sieved through a mesh screen. Fifty grams (50 g) of sieved sand was transferred into a 90-mm diameter Petri dish. Five mL of each fungal isolate (21 isolates) stock suspension, titrated to test concentrations of 1×10^7 and 1×10^8 conidia mL⁻¹ (for fruit flies) and 1×10^7 conidia mL⁻¹ for *T. leucotreta* were used to inoculate 50 g of sand in a single Petri dish, this was then vigorously mixed with a spatula. Freshly prepared, conidial suspension was used per replicate. The moisture level in the Petri dishes, after being treated with fungal suspension, was not determined however fungal treated sand did clump slightly. Controls were treated with sterilized distilled water containing 0.05% Triton X-100. Twenty *T. leucotreta*, *C. capitata*, and *C. rosa* final instar larvae, which would be ready to pupate within the next 24 h, were introduced into a single Petri dish, this constituting a replicate. Four replicate Petri dishes each with 20 insects was used per fungal isolate. The dishes were incubated at 25°C under a 12 h L:12 h D photoperiod. Seven days after the introduction of larvae, pupae were removed from Petri dishes containing treated sand and transferred into new Petri dishes containing sterilized, untreated sand. Sterilized plastic bottles were cut 15 cm below the neck of the bottle and placed over a replicate Petri dish (this acted as an emergence chamber and allowed emerging flies and moths the freedom to move around). The replicated Petri dishes covered with plastic emergence chambers were placed in trays filled with autoclaved Pearlite. For fruit flies, a cotton bud soaked in water and a small lid (3 × 3 cm) containing a 4:1 mixture of sugar and yeast hydrolysate, were introduced into the neck of the bottle and the emergence chamber respectively to provide food. For *T. leucotreta* adults, an 8% sugar solution was provided. Test insects were maintained at the same conditions as described above. The number of adult flies and moths that emerged from treated and control sand were recorded every second day until 10 days after the first emergence. Records were also kept of the number of pupae that failed to emerge. Adult flies and moths that died during this period and pupae or larvae that failed to emerge were surface sterilized in 70% ethanol and transferred onto SDA. The criteria for scoring mycoses were (1) failure of pupae to emerge followed by fungal sporulation and (2) death of adults followed by fungal sporulation.

2.5 Concentration–response of *T. leucotreta* to entomopathogenic fungi

Four fungal isolates that were the most virulent to all three pest insect species in previous bioassays were selected to assess the concentration–mortality relationship and to determine the lethal concentration LC₅₀ value for each fungal isolate against the citrus pest species, *T. leucotreta*. The registered microbial control product, BroadBand[®] was also included as an additional isolate in the

concentration–mortality trials. BroadBand[®] was supplied as a wettable powder formulation with a concentration of 2×10^{10} conidia/g of active *B. bassiana*. This product was stored and used according to the manufacturer's instructions. Conidial concentrations of the five isolates were determined using a Neubauer haemocytometer following 5-fold serial dilution in sterile distilled water supplemented with 0.05% Triton X-100. Based on previous bioassays, four treatments of fungal inocula were calibrated for these concentration–mortality relationship bioassays with *T. leucotreta*: 1×10^4 , 1×10^5 , 1×10^6 , 1×10^7 conidia mL⁻¹. Five mL of serial diluted conidial suspension was used to inoculate 50 g of autoclaved dried river sand in a single Petri dish. Twenty late instar, non-migrating, *T. leucotreta* larvae were placed into a single Petri dish, this constituted one replicate. Four replicate Petri dishes each containing 20 insects was used per treatment per isolate. A fresh colony scraped, newly titrated, conidial stock suspension was used per replicate and all multiple dose assays were conducted at the same time. Viability of conidia was determined and criteria for scoring mycosis were undertaken as described above.

2.6 Statistical analyses

Thaumatotibia leucotreta final instar larvae treated with entomopathogenic fungi did not always pupate normally because larvae were sometimes found mycosed. For the purpose of analyses, those mycosed larvae were grouped with mycosed pupae and considered as one grouped life stage. Few *T. leucotreta* adults were found mycosed and for this reason only adult emergence and not adult mycosis was recorded. In both fruit fly species, *C. capitata* and *C. rosa*, all final instar larvae treated with fungi pupated normally but infection was recorded in puparia and emerging adults. For *T. leucotreta* the percentage mortality data were corrected for control mortality (Abbott 1925) and the data were normalized through arcsine transformation and then subjected to analysis of variance (ANOVA) followed by mean separation by the Student–Newman–Keul's test ($P = 0.05$). For *C. capitata* and *C. rosa* the percentage mortality data were corrected for control mortality (Abbott 1925) and the data were normalized through square root transformation and then subjected to analysis of variance (ANOVA) followed by mean separation by the Holm–Sidak test ($P = 0.001$). To investigate whether fungal isolates from orchards or natural habitats showed grouping in terms of pathogenicity, this mortality data was normalized through arcsine transformation and subjected to ANOVA followed by mean separation by the Student–Newman–Keul's test ($P = 0.05$). Regression analysis was used to determine the functional relationship between log concentration of inoculum and probit of mortality (pupae and adults mycosis combined) using PROBAN at a test level of $P = 0.05$ (Van Ark 1995).

3. Results

3.1 The effect of lower concentrations of entomopathogenic fungi on *Ceratitis* spp. adult and pupal mycosis

In fungal conidial viability tests, germination for all isolates ranged from 88 to 95%. *Ceratitis rosa* adults and puparia were exposed to two fungal concentrations, 1×10^7 conidia mL⁻¹ and 1×10^8 conidia mL⁻¹. When undertaking bioassays at both

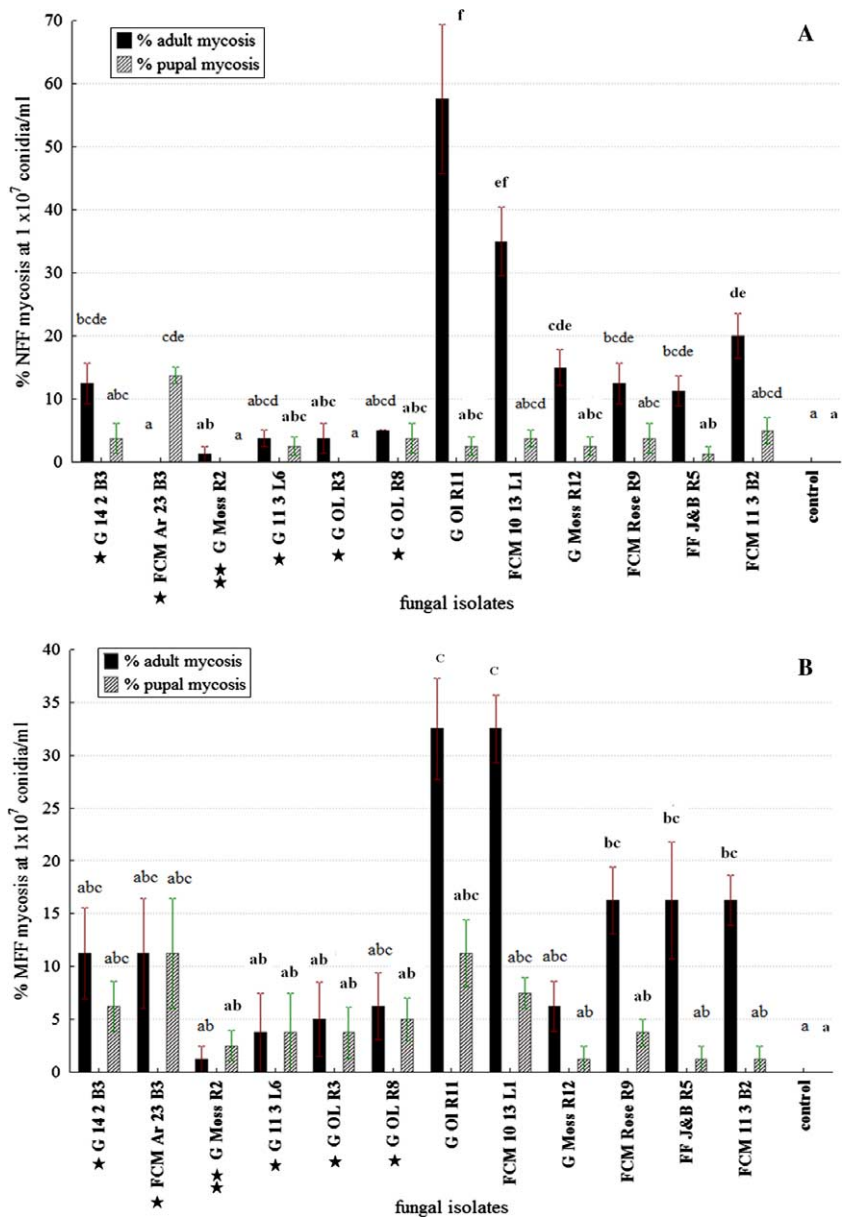


Figure 1. Pathogenicity of native fungal isolates of *B. bassiana*, *M. anisopliae* (denoted with a star), and *M. flavoviride* (two stars) to (A) *C. rosa*: Mean ($\% \pm \text{SE}$) of *C. rosa* adult and pupal mycosis after treatment with a concentration of 1×10^7 conidia mL^{-1} . (B) Mean ($\% \pm \text{SE}$) of *C. capitata* adult and pupal mycosis after treatment with a concentration of 1×10^7 conidia mL^{-1} . Bars with the same letter do not differ significantly by Holm-Sidak test ($P = 0.001$).

concentrations natural mortality in the untreated controls never exceeded 10%. When treating *C. rosa* final instar larvae with a lower fungal concentration, the percentage of puparia with visible signs of mycosis ranged from 1 to 14% depending on the fungal isolate (Figure 1A). The *M. anisopliae* isolate, FCM Ar 23 B3, had the

greatest effect on *C. rosa* puparia at this concentration. Deferred mortality due to mycosis in *C. rosa* adult flies ranged from 1 to 58% depending on the fungal isolate (Figure 1A). There was a large amount of variation among fungal isolates in their ability to suppress *C. rosa* adults and puparia ($F_{12,94} = 22.9$; $P = 0.001$) with two *B. bassiana* isolates, *G OL R11* and *FCM 10 13 LI*, showing a significant effect on *C. rosa* adults in particular. Entomopathogenic fungal isolates had a significantly greater effect on the adults of *C. rosa* than they did on the puparia of this fruit fly species ($F_{1,94} = 4.63$; $P = 0.034$), however this was a weak association. Further, the interaction between insect life stage (puparia and adults) and fungal isolate was weak ($F_{12,94} = 3.1$; $P = 0.001$) (Figure 1A).

The larvae of the second fruit fly species, *C. capitata*, were also treated with the same fungal concentrations as *C. rosa*. Mortality in the untreated controls never exceeded 10%. The use of the lower concentration, 1×10^7 conidia mL^{-1} , resulted in the percentage of puparia with visible signs of mycosis, ranging from 1 to 11% depending on fungal isolate (Figure 1B). The *M. anisopliae* isolate, *FCM Ar23 B3*, and *B. bassiana* isolate, *G OL R11*, had the greatest effect on the puparia of *C. capitata*. Deferred mortality in *C. capitata* adult flies ranged from 1 to 33% depending on fungal isolate (Figure 1B), with the same two *B. bassiana* isolates, *G OL R11* and *FCM 10 13 LI*, exhibiting the greatest pathogenicity towards *C. capitata* adults, as was seen with *C. rosa* adults at the same concentration. Fungal isolates had a significantly greater effect on the adults of *C. capitata* than on the puparia of this fruit fly species ($F_{1,94} = 23.61$; $P = 0.001$). Two *B. bassiana* isolates, *G OL R11* and *FCM 10 13 LI*, had a great effect on *C. capitata* than did the weaker *Metarhizium* isolates ($F_{12,94} = 11.89$; $P = 0.001$). There was however a significant but weak interaction between fruit fly life stage and fungal isolate ($F_{12,94} = 3$; $P = 0.001$) (Figure 1B).

3.2 The effect of higher concentrations of entomopathogenic fungi on *Ceratitis* spp. adult and pupal mycosis

The percentage of *C. rosa* puparia with visible signs of mycosis when treated with a concentration of 1×10^8 conidia mL^{-1} ranged from 1 to 25% with a *B. bassiana* isolate, *G 14 2 B5*, inducing the greatest percentage mycosis in puparia (Figure 2A). Deferred mortality in *C. rosa* adult flies ranged from 3 to 32% depending on the fungal isolate (Figure 2A). Puparia and adults varied significantly in response to fungi ($F_{1,80} = 9.81$; $P = 0.002$) with more adults responding to the fungal treatments than puparia however this again was weak. Four *B. bassiana* isolates, *G14 2 B5*, *G B Ar23 B3*, *FCM 10 13 LI*, and *G Moss R10* were more pathogenic and affected *C. rosa* life stages significantly compared to some weaker isolates, such as the two *M. anisopliae* isolates (*G 14 2 B3* & *FCM Ar 23 B3*) ($F_{11,80} = 21.61$; $P = 0.001$). Further, the interaction between insect life stage and isolate was significant but weak ($F_{11,80} = 2$; $P = 0.039$) (Figure 2A). Although percentage mycosis did increase with an increase in fungal concentration, a 10-fold dose-response was not observed in *C. rosa* puparia or adults.

The percentage of *C. capitata* puparia with visible signs of mycosis remained in the same range at the higher concentration as the lower concentration, from 1 to 10% (Figure 2B). Deferred mortality in *C. capitata* adults ranged from 4 to 41% depending on the fungal isolate (Figure 2B). Puparia and adults varied significantly

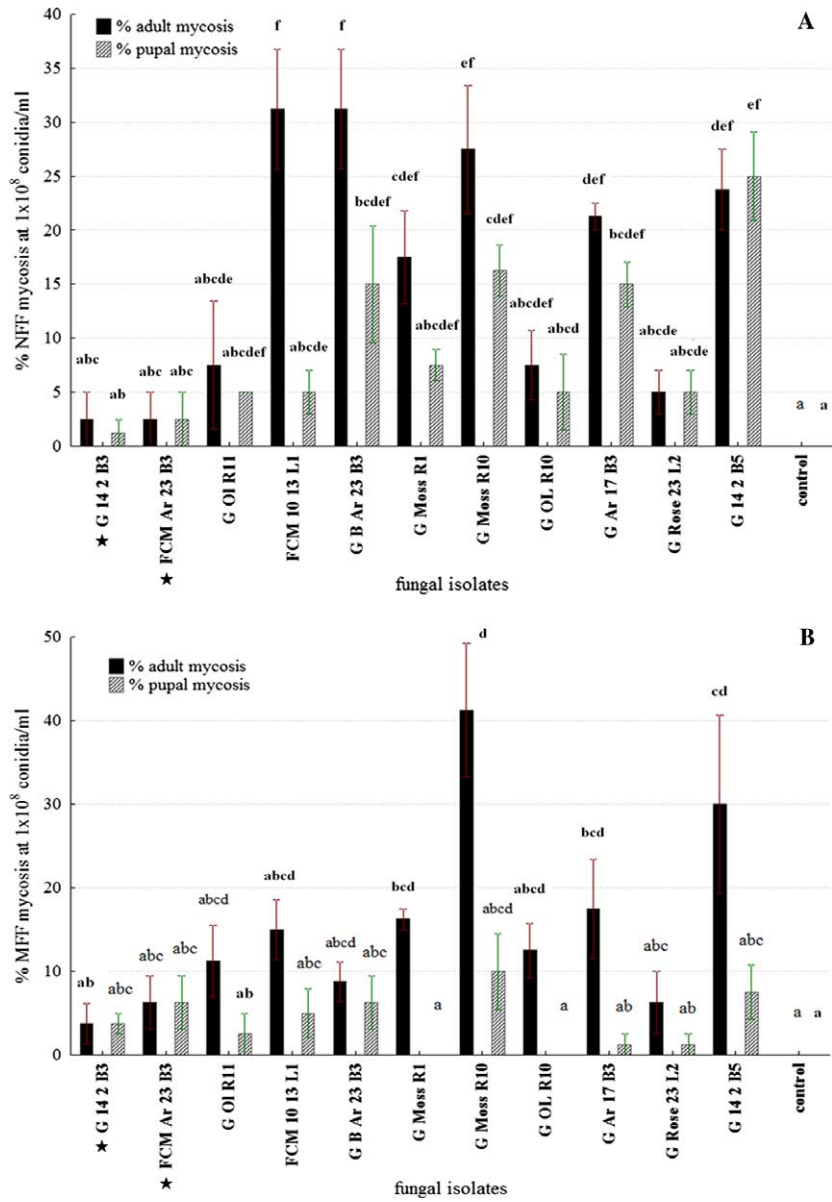


Figure 2. Pathogenicity of native fungal isolates of *B. bassiana* and *M. anisopliae* (denoted with a star) to (A) *C. rosa*: Mean (% $\pm$ SE) of *C. rosa* adult and pupal mycosis after treatment with a concentration of 1×10^8 conidia mL⁻¹. (B) Mean (% $\pm$ SE) of *C. capitata* adult and pupal mycosis after treatment with a concentration of 1×10^8 conidia mL⁻¹. Bars with the same letter do not differ significantly by Holm-Sidak test ($P = 0.001$).

in response to fungi ($F_{1,80} = 49.19$; $P = 0.001$) with a larger proportion of adults than puparia responding to the fungal treatments (Figure 2B). The two *B. bassiana* isolates, *G Moss R10* and *G 14 2 B5*, showed the greater pathogenicity towards *C. capitata* than did the other native isolates ($F_{11,80} = 8.13$; $P = 0.001$). Again, the

interaction between insect life stage and isolate was significant but weak ($F_{11,80} = 3.28$; $P = 0.001$) (Figure 2B). Percentage mycosis did not increase with an increase in fungal concentration and a 10-fold expected dose–response was not observed in *C. capitata* puparia or adults.

In a previous study (Goble et al. 2010) native entomopathogenic fungal isolates were collected from citrus orchards and adjacent natural habitats (refugia) (Table 1). It was of interest to ascertain whether grouping isolates from these habitats would reveal a pattern in pathogenicity. For example, ‘*Beauveria* diversity was shown to be highest in the seminatural habitats of a hedgerow in Denmark, which was likely attributed to the greater diversity and abundance of insect hosts in these areas compared to the structurally simpler agricultural field’ (Meyling et al. 2009). Thus the chance incidence of finding fungal isolates from natural habitats which may potentially be more pathogenic than isolates obtained from cultivated environments is likely. The effect of isolate habitat (either citrus orchards or refugia) on the incidence of fruit fly mycosis (both *C. capitata* and *C. rosa*) at a conidial concentration of 1×10^7 conidia mL^{-1} , showed that there were no significant differences ($F_{1,20} = 0.02$; $P = 0.87$) observed in the incidence of mycosis by various fungal isolates grouped according to habitat; the same was true of the higher dose concentration (1×10^8 conidia mL^{-1}) ($F_{1,18} = 0.07$; $P = 0.78$).

3.3 The effects of entomopathogenic fungi on *T. leucotreta* adult emergence and pupal mycosis

In fungal conidial viability tests, germination for all isolates ranged from 88 to 95%. When treating *T. leucotreta* larvae with a standard fungal concentration of 1×10^7 conidia mL^{-1} , 100% of adults emerged in the control. The treatments varied from 5 to 60% adult emergence, depending on fungal isolate ($F_{21,83} = 33.295$; $P = 0.001$) (Figure 3). Twelve of the 21 fungal isolates reduced *T. leucotreta* adult emergence to below 20%.

When *T. leucotreta* pupae were treated with various fungal isolates at a concentration of 1×10^7 conidia mL^{-1} , visible signs of mycosis on pupae ranged from 21 to 93% depending on the fungal isolate ($F_{21,83} = 96.436$; $P = 0.001$) (Figure 4). Notably, the same four *B. bassiana* isolates, FCM 10 13 L1, G B Ar 23 B3, G Moss R10, and G 14 2 B5 which showed pathogenicity towards fruit flies showed pathogenicity towards *T. leucotreta*, as seen by the decrease in adult emergence and an increase in percentage pupal mycosis with these isolates (Figure 4).

3.4 The effect of fungal concentration on *T. leucotreta* mortality

In fungal viability tests, germination of conidia for *B. bassiana* isolates, G 14 2 B5, G Moss R10, G B Ar 23 B3, FCM 10 13 L1 and BroadBand® ranged from 80 to 92%. The relative virulences of the five fungal isolates were compared against one another as log-probit regressions of mortality against *T. leucotreta* (Figure 5). When the five fungal regression lines were compared against one another a positive relationship between increased conidial concentration and *T. leucotreta* mortality was observed. When the regression lines were compared, residual variances were homogenous ($\chi^2 = 0.568$; $\text{df} = 4$; $P = 0.96$) and the slopes of the lines were compared. The slopes of the lines were found to be parallel ($\chi^2 = 2.821$; $\text{df} = 4$; $P = 0.59$) and the elevations

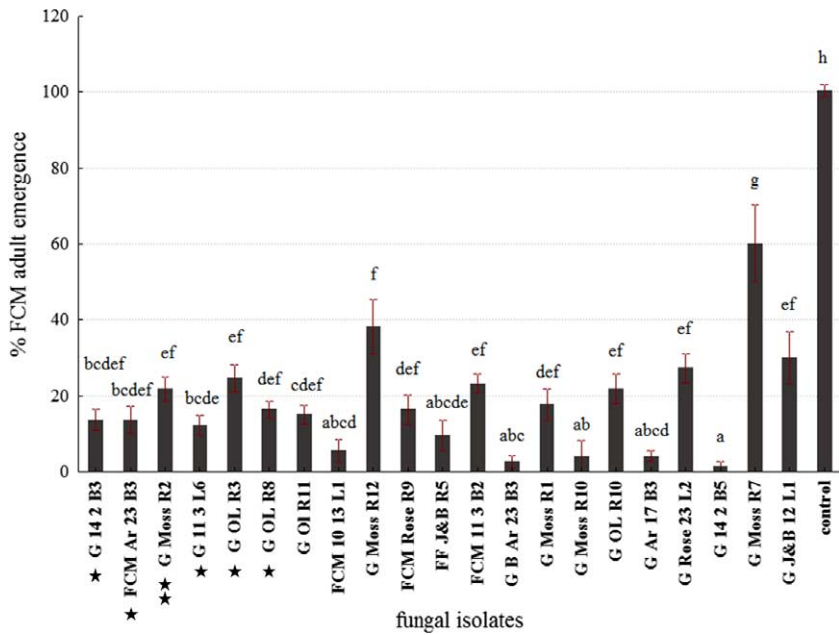


Figure 3. Pathogenicity of isolates of *B. bassiana*, *M. anisopliae* (denoted with a star), and *M. flavoviride* (two stars) to *T. leucotreta*: Mean (% ± SE) of adult emergence after treatment with a concentration of 1×10^7 conidia mL⁻¹. Bars with the same letter do not differ significantly by SNK test ($P = 0.05$).

were compared. The elevations of the lines were significantly different ($F_{4, 14} = 28.11$; $P = 0.0001$) and when multiple comparisons of the elevation and the slopes of the lines was undertaken using Bonferroni's test, mean separation ($P = 0.005$) showed that the isolates *G Moss R10*, *G 14 2 B5* and BroadBand[®] were significantly more virulent than *G B Ar 23 B3* and *FCM 10 13 L1* (Figure 5). These differences were observed in the relative LC₅₀ and LC₉₀ values of each isolate which are shown in Table 2. The slopes of the regression lines for isolates: *G Moss R10* and *G 14 2 B5* were steep relative to the other fungal isolates, indicating a stronger concentration response in *T. leucotreta* to these isolates, than to BroadBand[®]. An even weaker concentration response was observed against isolates: *G B Ar 23 B3* and *FCM 10 13 L1* (Figure 5).

4. Discussion

The native isolates of entomopathogenic fungi used in this study were not able to induce high mortality in the subterranean life stages of *C. capitata* and *C. rosa* when they were exposed to sand treated with conidia suspensions in contrast to other studies (Castillo, Moya, Hernandez, and Primo-Yufera 2000; Ekesi et al. 2002). During the initial fungal virulence screening trials, pupal mycosis of both fruit fly species did not exceed 25% by any fungal isolate, even at a higher dose concentration. Toledo et al. (2006), found the same to be true in a previous investigation of various *B. bassiana* and *M. anisopliae* isolates and formulated products against *Anastrepha*

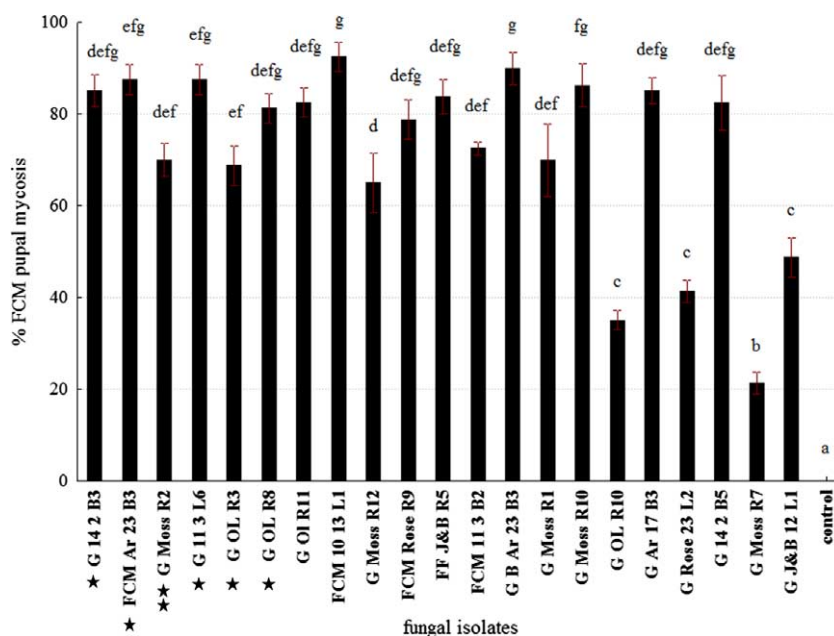


Figure 4. Pathogenicity of *B. bassiana*, *M. anisopliae* (denoted with a star), and *M. flavoviride* (two stars) to *T. leucotreta*: Mean ($\% \pm \text{SE}$) of pupal mycosis after treatment with a concentration of 1×10^7 conidia mL^{-1} . Bars with the same letter do not differ significantly by SNK test ($P = 0.05$).

ludens and *C. capitata*. They showed that entomopathogenic fungi had no effect on the larvae or pupae of these fruit fly species but that all isolates were highly virulent towards the adults. Similarly, De la Rosa et al. (2002) reported that the mortality caused by eight entomopathogenic fungal isolates at a concentration of 1.6×10^8 conidia mL^{-1} applied by dipping the immature stages of Mexican fruit fly, *A. ludens*, was low, 2–8% in larvae and none in pupae. However, very high levels of adult mortality were noted against three *B. bassiana* isolates which caused 98–100% mycosis. In the evaluation of susceptibility of *Delia antiqua* (Diptera: Anthomyiidae) to 11 isolates of hyphomycetes, pupae were generally found not to be susceptible to mycosis (Poprawski, Robert, Majchrowicz, and Bovin 1985). Ekesi et al. (2002) noted that when third instar fruit fly larvae enter fungus treated sand and pupate, the puparia may become sclerotized before fungal conidia can germinate to penetrate the pupal case. This may give some insight into why the immature life stages of fruit flies are less susceptible to pathogenic fungi than the adults.

Our results showed large variation among fungal isolates, at different concentrations, in their ability to suppress *C. rosa* and *C. capitata* adults and puparia. Further, there was a large amount of variability in susceptibility of fruit flies to fungal pathogens. Fungal isolates which showed marked pathogenicity towards fruit fly adults and puparia at the lower concentration (*G OL R11*, *G 14 2 B3*, and *FCM Ar 23 B3*) did not show the same virulence at a higher concentration. Although percentage mycosis did increase with an increase in fungal concentration in some instances, an expected 10-fold dose–response was not observed. Test insects selected

Table 2. The calculated probit data for each of the five *B. bassiana* isolates tested against *T. leucotreta* using the program PROBAN.

Insect species	Isolate	Fit of the line			Intercept	R-coefficient	Fiducial	Lethal concentration			
		χ^2	df	<i>P</i>				LC ₅₀	LC ₅₀ (SE)	LC ₉₀	LC ₉₀ (SE)
<i>T. leucotreta</i>	G Moss R10	2.44	2	0.29	3.19	0.47	0.12	6.8×10^5	4.8±	3.6×10^8	2.5±
	G 14 2 B5	1.86	2	0.39	3.17	0.46	0.11	8.8×10^5	5.9±	5.2×10^8	3.8±
	Broad	0.26	2	0.87	3.50	0.34	0.17	2.1×10^6	1.6±	1.1×10^{10} *	1.5±
	Band®										
	G B Ar 23 B3	0.73	2	0.69	3.26	0.33	0.18	1.6×10^7	9.4±	1.1×10^{11} *	2.2±
	FCM 10 13	0.64	2	0.73	2.93	0.37	0.15	3.7×10^7	1.8±	1.0×10^{11} *	1.8±
	L1										

*Data have been extrapolated due to insufficient points.

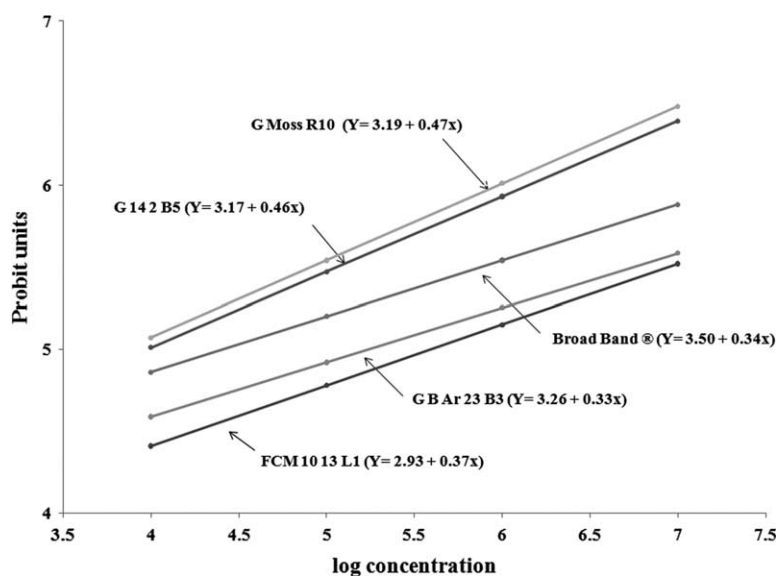


Figure 5. The log-probit regressions of mortality (pupal and adult combined) caused by five isolates of *B. bassiana* against *T. leucotreta*.

as late instars, approaching molt as in the case here, would be highly variable in their susceptibility to fungal pathogens, as molting removes the inoculum from the insect body (Begemann 2008). Further, fungal stock solutions were all titrated to the same concentration however different stock solutions were taken from colonies grown on different media plates which were then used to inoculate individual replicates. This may have further contributed to the observed variance.

Post emergence mycosis occurred in fruit fly adults escaping infection as puparia in all fungal screening trials. Puparia and adults varied significantly in response to fungi with a larger proportion of adults responding to the fungal treatments than the puparia. It is believed that the design of the bioassays (test insects were held together in groups of 20 for emergence) may have inflated the adult mortality data due to secondary infection. When young adult fruit flies infected with fungi died in the emergence chambers early on in the 10-day bioassay trial period, they became a source of infection for healthy adults when contact was made. In a study by Dimbi, Maniania, Lux, and Mueke (2003), the age of the host, in this case, *C. capitata*, had a significant effect on mortality, with younger flies (0–7 days old) succumbing to infection by entomopathogenic fungi earlier than older flies (7–14 days). If the chance incidence of young fruit flies with deferred fungal infection is experienced early on in the bioassay trial period, the incidence of mycosis in adult flies could become higher because of unquantified secondary infection.

Further, because of the duration (10-day) of the bioassay, adult fruit flies of both species were observed mating and eggs were also found on the cotton wool plugs, which were used to introduce water into the emergence chambers. Horizontal transmission of unquantified fungal inoculum between treated flies and healthy flies would have also caused additional variation. Thomas et al. (2001) reported that *C. capitata* females were able to oviposit as early as 4–5 days after emergence which

would explain why eggs were noted on the cotton wool bungs. After evaluating the SDA plates when the trials had terminated, both male and female *C. rosa* and *C. capitata* adults had entomopathogenic fungi growing on the tip of the abdomen and on the thoracic sclerites, indicating that horizontal transmission had occurred. Quesada-Moraga, Martin-Carballo, Garrido-Jurado, and Santiago-Álvarez (2008) reported that when uninfected *C. capitata* females were paired with *M. anisopliae* inoculated males for 24 h, these females showed a higher density of conidia in the dorsal thoracic sclerites. Inoculated males exhibited conidia on the aedeagi and females had conidia on the ovipositor (Quesada-Moraga et al. 2008). Horizontal transmission of fungi in the laboratory has also been noted in diamondback moth, *Plutella xylostella* (Furlong and Pell 2001) and in the malaria vector, *Anopheles gambiae* (Scholte, Knols, and Takken 2004). Perhaps earlier termination of the emergence stage of the bioassays would have lowered the chances of horizontal transmission and secondary inoculum pick up from cadavers, lowering adult mortality data. Future studies should consider very carefully their post emergence bioassay structure to avoid misleading data. However, adult fruit flies were found to be more susceptible to the effects of fungal pathogens than the puparia and this has been noted by other authors (De la Rosa et al. 2002; Ekesi et al. 2002; Toledo et al. 2006).

Most of the native isolates of fungi which were tested in this study were able to induce mortality in *T. leucotreta* final instar larvae and pupae when exposed to sand treated with a conidial suspension. The overall effect of this subterranean treatment led to a significant reduction in adult moth emergence (less than 20% emergence was achieved with most fungal isolates). Eleven of the initial 21 fungal isolates tested showed pathogenicity towards the pupae (mycosis greater than 80%) of this susceptible insect species. Concentration–response trials against *T. leucotreta* revealed a dose-dependent relationship and two selected isolates, *G 14 2 B5* and *G Moss R10* could induce mycosis in 50% of the treated population (LC_{50}) at concentrations of 8.8×10^5 and 6.8×10^5 conidia mL^{-1} , respectively. Similarly, in a study by Begemann (2008), various concentrations of *B. bassiana* in suspension and as dry preparations were tested against the final instar, non-migrating larvae of *T. leucotreta*. The author found that non-migrating larvae illustrated a dose-dependent response with LC_{50} and LC_{90} values in the range of 5.2×10^4 to 1.0×10^6 and were more susceptible to the fungus. There was however no dose-dependent response observed for migrating *T. leucotreta* larvae and this was attributed to lowered susceptibility, possibly due to the molting process already having occurred.

Anand, Prasad, and Tiwary (2009) reported that entomopathogenic fungi applied to *Spodoptera litura* (Lepidoptera: Noctuidae) pupae in the soil, as a drench (conidial suspensions at a titre of 10^8 conidia g^{-1} of soil) reduced adult emergence by up to 80%, depending on the fungal isolate. However, when premixing the conidial suspensions into the soil, the fungi killed fewer pupae; this was attributed to the accumulation of conidia up to 10 cm from the soil surface during drenching. Our study revealed effective results against *T. leucotreta* using pre-mixed conidial suspensions because larvae move continuously throughout the soil before they pupate. Anand et al. (2009) reported LC_{50} and LC_{90} values of *M. anisopliae* (the most virulent isolate) as 1.2×10^7 and 4.3×10^8 conidia g^{-1} , respectively. In the investigation of entomopathogenic fungi against *Thaumetopoea pityocampa* (Lepidoptera: Thaumetopoeidae) at concentrations of 1×10^5 conidia mL^{-1} , all isolates

tested resulted in various levels of mortality (16–100%) and when dose–mortality tests were undertaken by spraying five conidial concentrations on larvae, the estimated LC_{50} value was 3.4×10^6 conidia mL^{-1} (Er, Tunaz, and Gokçe 2007). Loc and Chi (2007) reported that *B. bassiana*, *M. anisopliae* and two commercially available biopesticides (Crymax® 35 WP and Atabron® 5EC) applied at concentrations of 1×10^7 conidia mL^{-1} were able to induce 70–87.3% mortality in, *P. xylostella*, larvae 7 days after exposure when applied directly as sprays. Tefera and Pringle (2003) reported high mortality (98–100%) in the stem borer, *Chilo partellus*, (Lepidoptera: Pyralidae) after exposing second instar larvae to *B. bassiana* at concentrations of 1.25×10^7 conidia mL^{-1} . Compared to the above mentioned studies it appears that *T. leucotreta* has a ‘within range’ susceptibility to the effects of native South African entomopathogenic fungi. No particular isolate was highly virulent towards this insect species however the two *B. bassiana* isolates are considered effective candidates as a platform for future study.

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